Second-Generation Hexavalent Molybdenum Oxo-Amidinate Precursors for Atomic Layer Deposition

Titel Jurca,[a] Aaron W. Peters,[a] Aidan R. Mouat,[a] Omar K. Farha,[a] Joseph T. Hupp,[a] Tracy L. Lohr,[a] Massimiliano Delferro,[a,b]* and Tobin J. Marks[a,]*

[a]Department of Chemistry and the Materials Research Center, Northwestern University, Evanston, Illinois 60208-3113, USA

[b]Chemical Sciences and Engineering Division, Argonne National Laboratory, Lemont, Illinois 60439, USA

Supporting Information Placeholder

ABSTRACT: The synthesis of molybdenum oxo-amidinate complexes $MoO_2(R\text{-}AMD)_2$ [AMD = N,N'-di-R-acetamidinate; R = Cy (**2;** cyclohexyl) and ^{i}Pr (**3**)], and their characterization by ^1H, ^{13}C NMR spectroscopy, single-crystal X-ray diffraction, and thermogravimetric analysis is reported. Quartz-crystal microbalance and film X-ray photoelectron spectroscopic studies confirm that **3** is an improved second-generation ALD precursor for MoO_3 film growth which can be synthesized in high yields (80%+), is easier to handle without mass loss, and in conjunction with O_3 as the second ALD reagent, yields nitride-free MoO_3 films.

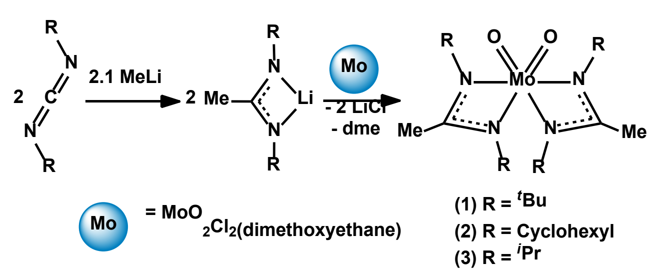
As a result of the multiple accessible molybdenum oxidation states, and an anisotropic layered crystal structure, MoO_3 has garnered widespread application in fields ranging from gas sensing,[1] energy storage,[2] field emission,[3] photo- and electrochromic devices,[4,5] hole transport layers for inverted architecture organic photovoltaics (OPVs),[6-8] and catalysis. Notably, owing to their wide range of technologically-relevant applications as heterogeneous catalysts for transformations such as hydrocracking,[9] hydrodeoxygenation,[10] hydrodesulferization,[11] olefin polymerization,[12] and olefin metathesis,[13] supported molybdenum oxide (MoO_x) catalysts are of significant interest. The broad-ranging applicability of MoO_x catalysts is attributed to Brønsted or Lewis acid properties, and/or offering redox-active sites.[14] Although the definitive structure of the active sites in these catalysts is still under investigation, the presence of molybdenum-oxo functionalities in either monomeric or oligomeric/polymeric surface species is broadly implicated.[15]

Catalytic activity and selectivity are known to vary as a function of MoO_x surface density, with oligomeric and polymeric species exhibiting the most desirable properties.[16,17] However, increased loadings (greater than a monolayer) have been shown to promote crystalline domain growth, which reduces the density of single active sites, and ultimately erodes the catalytic activity.[18] The structure of supported MoO_x, and related catalytic activity and selectivity, is also significantly influenced by the synthetic approaches utilized.[19] As such, preparative methods for supported MoO_x species which afford precise control over the resulting surface structures would be highly desirable.

Atomic layer deposition (ALD) is a powerful film deposition technique that enables uniform conformal film growth on high aspect ratio and surface area materials with Ångstrom-level thickness control.[20] Thus ALD has recently attracted great interest as a method for the precise fabrication of heterogeneous catalysts.[21-27] In a typical ALD process, this is achieved by alternate pulsing of vapor-phase metal precursor and a gaseous oxygen or protonolytic reagent over an oxide support. Consumption of surface reactive sites, or steric crowding terminates the reaction, rendering the growth process self-limiting.[20] Importantly, continued advances in this promising field are contingent on developing new metal-based precursors displaying the necessary, yet scarce combined traits of volatility, thermal stability, and desired surface reactivity.

Recently, this Laboratory reported the synthesis, and subsequent ALD implementation of volatile oxo-amidinate complexes of Mo (**1**), and analogous W. Compound **1**, $MoO_2(^tBu\text{-}AMD)_2$ (^{t}BuAMD = N,N'-di-tert-butyl-acetamidinate) was accessed via a two-step synthesis (Scheme 1), and successfully utilized as a precursor for the ALD growth (employing O_3 as the second reagent) of partially nitrided molybdenum trioxide (MoON) ultrathin films (<10 nm, approximate composition $MoO_{2.48}N_{0.18}$).[28] Nevertheless, complex **1** is difficult to purify and is typically obtained as mixtures of semi-crystalline solid and highly viscous oil. Such products are difficult to manipulate, and **1** is obtained in generally low and variable isolated yields (20-63%).

Scheme 1. Synthesis of $MoO_2(R\text{-}AMD)_2$ ALD Precursors, R = ^{t}Bu (1), Cy (2), ^{i}Pr (3).



With the aim of improving synthetic consistency, increasing yields, minimizing mass loss under routine handling, and enhancing MoO_x film homogeneity by suppressing nitridation, we sought to elaborate the range of molybdenum oxo-amidinate species available for use as ALD precursors. Here we report the synthesis, and characterization of $MoO_2(R\text{-}AMD)_2$ (AMD = N,N'-di-R-acetamidinate) complexes with R = Cy (**2**) and ^{i}Pr (**3**)), and their implementation in ALD growth of MoO_3 films.

Compounds **2** and **3** are readily prepared via the nucleophilic attack of MeLi on the respective N,N'-di-alkylcarbodiimides,[29,30] followed by salt metathesis of the lithiated salt with MoO_2Cl_2(dme) (Scheme 1). Purification is achieved either by vacuum sublimation or recrystallization; optimized yields of 60% (**2**) and 85% (**3**) are achieved by recrystallization from concentrated Et_2O solution at -30 ºC, and sublimation at 110 ˚C/10^{-2} Torr, respectively. Although this synthetic protocol necessitates air- and moisture-free techniques, isolated **2** and **3** are both air- and moisture-stable. Compounds **2** and **3** were characterized by ^{1}H and ^{13}C NMR, FTIR, and UV-Vis spectroscopy (*see Supporting Information*). Single crystals suitable for X-ray diffraction were grown from concentrated solutions of Et_2O (**2**) and hexanes (**3**) at -30ºC (Figure 1).

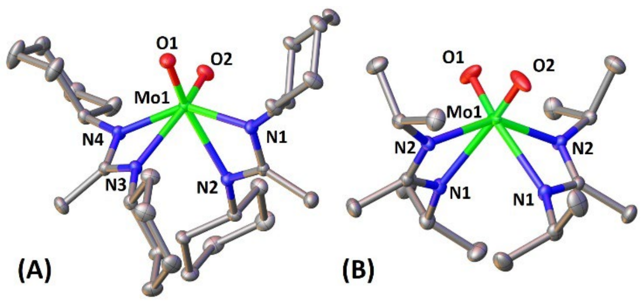


Figure 1. Solid state structures of precursors **2** (A) and **3** (B) with hydrogen atoms omitted for clarity, and ellipsoids drawn at the 50% probability level. Selected bond lengths [Å] and angles [°] **2**: Mo1-O1 1.708(3), Mo1-O2 1.707(3), Mo1-N1 2.082(3), Mo1-N2 2.282(3), Mo1-N3 2.309(3), Mo1-N4 2.085(3), ∠O1-Mo1-O2 105.98(14); **3**: Mo1-O1 1.7076(6), Mo1-N1 2.2878(5), Mo1-N2 2.0865(5), ∠O1-Mo1-O1 106.27(6).

As expected for heteroleptic complexes, *cis*-dioxo compounds **2** and **3** are six-coordinate, displaying significant distortion from an ideal octahedral geometry, and typical of Mo^{VI} *cis*-dioxo complexes.[28,31,32] The single crystal X-ray structures reveal propeller chirality, with **2** (and **1**) exhibiting a clockwise (Δ) and **3**, an anti-clockwise (Λ) twist (Figure S5). The O-Mo-O angles for **2** [105.98(14)˚] and **3** [106.27(6)˚] are in good agreement with those reported for **1** (105.2˚)[28] and the parent dioxo-dichloro-complex MoO_2Cl_2(dme) (104.5˚).[32] The bidentate nature of the amidinate ligands results in a bite angle < 60˚ for both complexes and inter-ligand N-Mo-N bond angles > 90˚. The Mo=O bond lengths in **2** [1.708(3) and 1.707(3) Å] and **3** [1.7076(6) Å] are similar to those previously reported for **1** (1.692 - 1.698 Å)[25] and MoO_2Cl_2(dme) (1.673 – 1.667 Å).[32] A *trans*-effect is evident in the Mo-N bonds *trans*- to the oxo-ligands, which are elongated by ca. 10% versus the undistorted Mo-N bond lengths; a similar effect is operative in **1**.[28] Average Mo-N bond lengths decrease from **1** to **3** [**1** = 2.205(4) Å; **2** = 2.190(4) Å; **3** = 2.187(6) Å], consistent with decreasing steric congestion imposed by the R group (^{t}Bu > Cy > ^{i}Pr).

While complex **3** possesses C_2 symmetry, both **2**, and **1** are C_1-symmetric. Although minimal, measurements of Mo distance from the calculated plane α (α = N, N, N, N, Mo, O, O) reveal a trend of increasing deviation from the higher C_2 symmetry of **3** in the order **3** (Mo – α = 0.000 Å) < **2** (– α = 0.007 Å) < **1** (Mo – α = 0.011 Å) (Figure S6). This trend of increasing asymmetry from **3** → **1** is paralleled in the respective crystal systems, where **3** = tetragonal, **2** = orthorhombic, and **1** = monoclinic.[28]

Thermogravimetric analysis (TGA) of complexes **2** and **3** performed in a glove box (Figures 2 and S6) indicates an onset of volatilization at ~120 ˚C. Both **2** and **3** exhibit smooth volatilization until 270 ˚C, and 260°C respectively, where inflection in the TGA curve indicates either a second volatilization or a decomposition event ending at 340°C for **2** and 310°C for **3**. The effect is significantly more pronounced for **2**. This phenomenom was also previously observed for **1**[28] and is currently under further investigation. The TGA data set the ALD window between 120 - 270°C for **2** and 120 - 260°C for **3**.

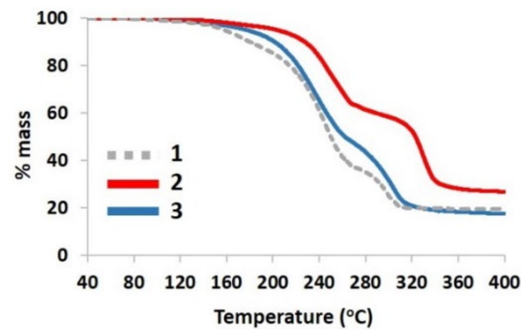


Figure 2. Air-free TGA analysis of **1**-**3**. Ramp rate 5 °C/min. under flow of N_2 (20 mL/min.).

The volatility of compounds **1**-**3** was compared relative to molecular weight (**1**: 466.54 g/mol; **2**: 570.69 g/mol; **3**: 410.43 g/mol) as a function of both temperature (°C) at 25% mass loss (**1**: 224 °C; **2**: 252 °C; **3**: 227 °C) and % mass loss at 250 °C (**1**: 50 %; **2**: 24 %; **3**: 44 %) (Figure S7). The decreased volatility of complex **2** versus **1** and **3** is expected due to the substantially higher molecular weight (22% and 39% higher respectively). However, the thermogravimetric properties of **1** and **3** are similar (within the ± 5 °C uncertainty of the TGA instrument), irrespective of the 14% higher mass of **1**. We attribute this to the increased asymmetry of **1** versus **3** (*vide supra*). Increased ligand, and consequently complex asymmetry, is known to promote volitalization.[33]

To test the efficacy of complexes **2** and **3** for ALD growth of MoO_3, we first conducted QCM studies utilizing O_3 in a conventional "A-B" cycle sequence (A = **2** or **3**, B = O_3) with a timing structure of t_1–t_2–t_3–t_4 (t_1 = pulse time of precursor; t_2 = purge time; t_3 = pulse time of O_3; t_4 = purge time after O_3 pulse). Attempted deposition with **2** at 150-225 °C proved unsuccessful. However, experiments with **3** and O_3 resulted in film growth. Optimized pulse times (t_1 and t_3) for **3** and O_3 were 2 s and 10 s, respectively and the purge times (t_2 and t_4) were 30 s and 45 s respectively. The QCM data (Figure 3) indicate a slow, but essentially linear rate of growth over 50 "A-B" cycles on the ALD-derived Al_2O_3 starting surface. During dosing of compound **3**, the average mass uptake $\mathbf{\Delta m_1}$ is 27.9 ± 3.9 ng/cm^2. Mass loss occurs during the introduction of O_3, suggesting that protonated ligand remains on the surface after deposition of compound **3**, which then reacts and desorbs with the introduction of O_3; $\mathbf{\Delta m_2}$ averages 23.6 ± 3.6 ng/cm^2. The average total mass gain per cycle ($\mathbf{\Delta m_3} = \mathbf{\Delta m_1} - \mathbf{\Delta m_2}$) is 4.3 ± 0.6 ng/cm^2, corresponding to a theoretical surface coverage rate of *ca*. 0.2 Mo/nm^2 per cycle (assuming predominantly MoO_3 layer composition). This number is approximate, however, as an unknown mass of decomposed ligand is also incorporated on the surface of the film (*vide infra*, *Figure 4*).

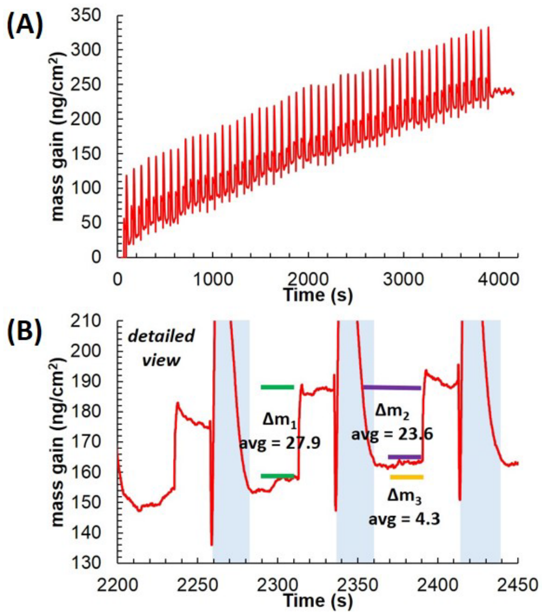


Figure 3. (A) QCM studies of the A-B type ALD growth process for MoO_3 films with precursor **3**. (B) detailed view; *blue highlights represent pressure spikes during O_3 pulses.*

Efforts to deposit MoO_3 (or MoON) with other B reagents (H_2O, H_2O_2, O_2) show initial growth, but this ceases after ~3 A-B cycles. Additionally, negligible growth is observed by *in-situ* QCM at high temperatures (>200 °C), confirming that **3** does not deposit via thermal decomposition/chemical vapor deposition. This agrees well with the TGA data suggesting that compound **3** is stable < 260 ˚C.

The morphology and microstructure of the **3** + O_3-derived films were first characterized by Atomic Force Microscopy (AFM) and X-ray diffraction (XRD). AFM images of the 50 cycle and 250 cycle MoO_3 growth process films (Figure S9) reveals smooth, continuous films and with RMS roughness = 0.227 nm and 0.251 nm, respectively. Θ-2θ XRD scans of both films yield featureless spectra, consistent with amorphous character.

To investigate the surface chemical composition of the ALD-derived films, and the Mo valence states Mo, X-ray photoelectron spectroscopy (XPS) was conducted on a Si (100) wafer coated with 50 ALD cycles of **3** + O_3. All binding energies are referenced to the adventitious C(1s) peak at 284.8 eV. Analysis of the resulting as-deposited films reveals expected N impurities; the main N(1s) peak at 398.9 eV corresponds to N impurities, but also contains contributions from an oxidized N species at 401.1 eV and the Mo($3p^3/_2$) peak at 394.5 eV (Figure 4).[34] The N(1s) peak at 401.1 eV is assigned to surface nitrate, and the peak at 398.9 eV to amine surface contaminants.[35,36] Analysis of the Mo(3d) region reveals peaks at 232.2 and 235.4 eV, which correspond to Mo(VI)($3d^5/_2$) and Mo(VI)($3d^3/_2$) respectively (Figure 4). This indicates that the films consisted entirely of Mo in the 6+ oxidation state, consistent with a predominant MoO_3 composition.

In contrast to the present results, partially nitrided films were previously obtained utilizing **1** and O_3 as the A-B reagents, and exhibited signals for both Mo(VI)($3d^5/_2$), Mo(VI)($3d^3/_2$) as well as Mo(IV)($3d^5/_2$) and Mo(IV)($3d^3/_2$). To confirm the nitridation is suppressed in the present growth experiments, with ultimate formation of MoO_3 when precursor **3** and O_3 are used, low intensity Ar^+ etching was conducted for 5 s to remove N-based surface contaminants from the as-grown films. This was sufficient to remove all traces of the N(1s) peaks (Figure 4, *denoted by* ♦). These etching conditions however result in the reduction of Mo(VI) to Mo(IV) as evidenced by disappearance of the Mo(VI) peaks and appearance of Mo(IV) peaks at lower binding energies (227.7 and 230.8 eV), a feature commonly observed on Ar^+ etching.[35]

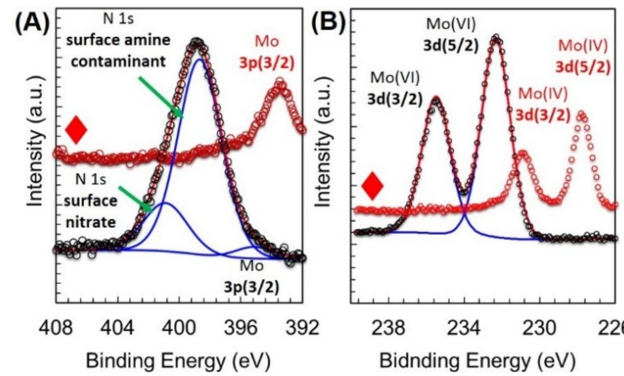


Figure 4. (A) N(1s)/Mo(3p) and (B) Mo(3d) XPS peaks in 50 cycle MoO_3 films from precursor **3** (♦ = *after etching with low intensity Ar ions for 5 s; Mo(VI) is reduced to Mo(IV)*).

In conclusion, we have expanded the scope of Mo oxo-amidinates as ALD precursors with complexes **2** and **3**. In particular, with **3**, we have developed a second-generation Mo oxo-amidinate ALD precursor which can be consistently accessed in high yields (80%+), is easy to handle with no mass loss, and maintains the thermogravimetric properties needed to facilitate ALD growth. Notably, films fabricated utilizing **3** and O_3 in an A-B cycle result in MoO_3 with only N-containing impurities on the surface. In contrast, similar processes utilizing **1** and O_3 resulted in partially nitrided molybdenum trioxide ($MoO_{2.48}N_{0.18}$) films.[28] Ongoing studies are focusing on further expanding this family of compounds, and detailed characterization of the resulting fabricated films.

ASSOCIATED CONTENT

Supporting Information. Full experimental details, including NMR and TGA characterization, and X-ray crystallographic data for **2** and **3**, and AFM micrographs of MoO_3 films. This material is available free of charge via the Internet at http://pubs.acs.org.

AUTHOR INFORMATION

Corresponding Authors

*E-mail: t-marks@northwestern.edu (T.J.M.)
*delferro@anl.gov (M.D.)

Funding Sources

Research was supported by the Chemical Sciences, Geosciences, and Biosciences Division, U.S. Department of Energy through a grant DE FG02-03ER15457 to the Institute of Catalysis in Energy Processes (ICEP) at Northwestern University.

Notes

The authors declare no competing financial interest.

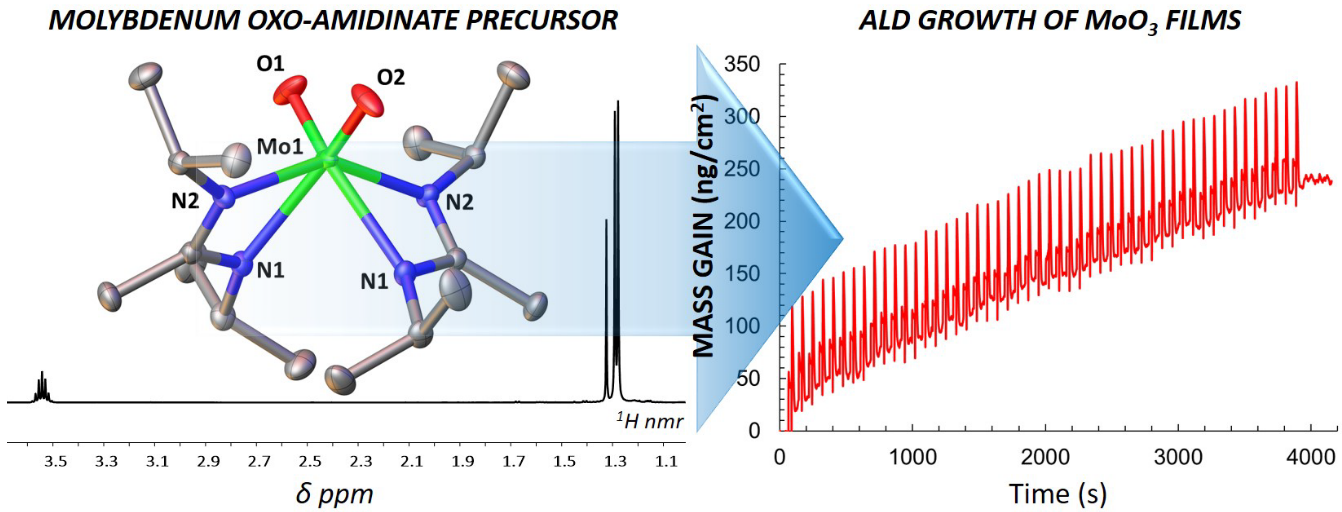
ACKNOWLEDGMENT

A.W.P. acknowledges support from the Department of Defense (DoD) through the National Defense Science & Engineering Graduate Fellowship (NDSEG) Program. W. Huang and Dr. A. Henning at Northwestern University are acknowledged for help with film characterization.

REFERENCES

1. Taurino, A. M.; Forleo, A.; Francioso, L.; Siciliano, P.; Stalder, M.; Nesper, R., Synthesis, electrical characterization, and gas sensing properties of molybdenum oxide nanorods. *Appl. Phys. Lett.* **2006**, *88*, 152111-3.
2. Liang, R.; Cao, H.; Qian, D., MoO_3 nanowires as electrochemical pseudocapacitor materials. *Chem. Commun,* **2011**, *47*, 10305-10307.
3. Zhou, J.; Xu, N.-S.; Deng, S.-Z.; Chen, J.; She, J.-C.; Wang, Z.-L., Large-Area Nanowire Arrays of Molybdenum and Molybdenum Oxides: Synthesis and Field Emission Properties, Adv. Mater. **2003**, *15*, 1835-1840.
4. Yao, J. N.; Hashimoto, K.; Fujishima, A., Photochromism induced in an electrolytically pretreated MoO_3 thin film by visible light, *Nature*, **1992**, *355*, 624-626.
5. Yang, Y. A.; Cao, Y. W.; Loo, B. H.; Yao, J. N., Microstructures of electrochromic MoO_3 thin films colored by injection of different cations. *J. Phys. Chem. B*. **1998**, *102*, 9392-9396.
6. Servaites, J. D.; Yeganeh, S.; Marks, T. J.; Ratner, M. A., Efficiency enhancement in organic photovoltaic cells: Consequences of optimizing series resistance. *Adv. Funct. Mater.* **2010**, *20*, 97-104.
7. Motiei, L.; Yao, Y.; Choudhury, J.; Marks, T. J.; van der Boom, M. E.; Facchetti, A., Self-propagating molecular assemblies as interlayers for efficient inverted bulk-heterojunction solar cells. *J. Am. Chem. Soc.*, **2010**, *132*, 12528-12530.
8. Zhou, N.; Guo, X.; Ortiz, R. P.; Li, S.; Zhang, S.; Chang, R. P. H.; Facchetti, A.; Marks, T. J., Bithiophene imide and benzodithiophene copolymers for efficient inverted polymer solar cells. *Adv. Mater.,* ,**2012**, *17*, 2242-2248.
9. Murrell, L. L.; Grenoble, D. C.; Kim, C. J.; Dispenziere Jr, N. C., Supported transition metal oxides as acid cracking catalysts: Periodic trends and their relationship to activity and selectivity. *J. Catal.* **1987**, *107*, 463-470.
10. Prasomsri, T.; Shetty, M.; Murugappan, K.; Roman-Leshkov, Y., Insights into the catalytic activity and surface modification of MoO_3 during the hydrodeoxygenation of lignin-derived model compounds into aromatic hydrocarbons under low hydrogen pressures. *Energ. Environ. Sci.* **2014**, *7*, 2660-2669.
11. Chianelli, R. R.; Daage, M.; Ledoux, M. J., Fundamental Studies of Transition-Metal Sulfide Catalytic Materials. *Advances in Catalysis*, D.D. Eley, H. P.; Werner, O. H., Eds. Academic Press: 1994; Vol. Volume 40, pp 177-232.
12. Cao, X.; Cheng, R.; Liu, Z.; Wang, L.; Dong, Q.; He, X.; Liu, B., DFT and PIO study of the influences of Mo valance state and surface hydroxyl on supported-MoOx catalysts for ethylene polymerization. *J. Mol. Catal. A: Chem.* **2010**, *321*, 50-60.
13. Lwin, S.; Wachs, I. E., Olefin metathesis by supported metal oxide catalysts. *ACS Catal.* **2014**, *4*, 2505-2520.
14. Amakawa, K.; Wrabetz, S.; Kröhnert, J.; Tzolova-Müller, G.; Schlögl, R.; Trunschke, A., In situ generation of active sites in olefin metathesis. *J. Am. Chem. Soc.* **2012**, *134*, 11462-11473.
15. Amakawa, K.; Sun, L.; Guo, C.; Hävecker, M.; Kube, P.; Wachs, I. E.; Lwin, S.; Frenkel, A. I.; Patlolla, A.; Hermann, K.; Schlögl, R.; Trunschke, A., How strain affects the reactivity of surface metal oxide catalysts. *Angew. Chem. Int. Ed.* **2013**, *52*, 13553-13557.
16. Caro, C.; Thirunavukkarasu, K.; Anilkumar, M.; Shiju, N. R.; Rothenberg, G., Selective Autooxidation of ethanol over titania-supported molybdenum oxide catalysts: structure and reactivity. *Adv. Synth. Catal.* **2012**, *354*, 1327-1336.
17. Liu, T.-C.; Forissier, M.; Coudurier, G.; Vedrine, J. C., Properties of molybdate species supported on silica. *J. Chem. Soc., Faraday Trans.* **1989**, *85*, 1607-1618.
18. Kitano, T.; Okazaki, S.; Shishido, T.; Teramura, K.; Tanaka, T., Brønsted acid generation of alumina-supported molybdenum oxide calcined at high temperatures: Characterization by acid-catalyzed reactions and spectroscopic methods. *J. Mol. Catal. A: Chem.* **2013**, *371*, 21-28.
19. Thielemann, J. P.; Hess, C., Monitoring silica supported molybdenum oxide catalysts at work: A Raman spectroscopic study. *ChemPhysChem* **2013**, *14*, 441-447.
20. George, S. M., Atomic layer deposition: An overview. *Chem. Rev.* **2010**, *110*, 111-131.
21. Feng, H.; Elam, J. W.; Libera, J. A.; Pellin, M. J.; Stair, P. C., Oxidative dehydrogenation of cyclohexane over alumina-supported vanadium oxide nanoliths. *J. Catal.* **2010**, *269*, 421-431.
22. Lu, J.; Fu, B.; Kung, M. C.; Xiao, G.; Elam, J. W.; Kung, H. H.; Stair, P. C., Coking- and sintering-resistant palladium catalysts achieved through atomic layer deposition. *Science* **2012**, *335*, 1205-1208.
23. O'Neill, B. J.; Jackson, D. H. K.; Lee, J.; Canlas, C.; Stair, P. C.; Marshall, C. L.; Elam, J. W.; Kuech, T. F.; Dumesic, J. A.; Huber, G. W., Catalyst design with atomic layer deposition. *ACS Catal.* **2015**, *5*, 1804-1825.
24. Lu, J.; Low, K.-B.; Lei, Y.; Libera, J. A.; Nicholls, A.; Stair, P. C.; Elam, J. W., Toward atomically-precise synthesis of supported bimetallic nanoparticles using atomic layer deposition. *Nat. Commun.* **2014**, *5*.
25. Kung, C. W.; Mondloch, J. E.; Wang; Wang, T. C.; Bury, W.; Hoffeditz, W.; Klahr, B. M.; Klet, R. C.; Pellin, M. J.; Farha, O. K.; Hupp, J. T., Metal-organic framework thin films as platforms for atomic layer deposition of cobalt ions to enable electrocatalytic water oxidation. *ACS Appl. Mater. Interfaces*, **2015**, *7*, 28223-28230.
26. Li, Z.; Schweitzer, N. M.; League, A. B.; Bernales, V.; Peters, A. W.; Getsoian, A.; Wang, T. C.; Miller, J. T.; Viunov, A.; Fulton, J. L.; Lercher, J. A.; Cramer, C. J.; Gagliardi, L.; Hupp, J. T.; Farha, O. K., Sintering-resistant single-site nickel catalyst supported by metal-organic framework. *J. Am. Chem. Soc.*, **2016**, *138*, 1977-1982.
27 Peters, A. W.; Li, Z.; Farha, O. K.; Hupp, J. T., Toward inexpensive photocatalytic hydrogen evolution: a nickel sulfide catalyst supported on a high-stability metal-organic framework. *ACS Appl. Mater. Interfaces*, **2016**, *8*, 20675-20681.
28. Mouat, A. R.; Mane, A. U.; Elam, J. W.; Delferro, M.; Marks, T. J.; Stair, P. C., Volatile hexavalent oxo-amidinate complexes: Molybdenum and tungsten precursors for atomic layer deposition. *Chem. Mater.* **2016**, *28*, 1907-1919.
29. Lim, B. S.; Rahtu, A.; Park, J.-S.; Gordon, R. G., Synthesis and characterization of volatile, thermally stable, reactive transition metal amidinates. *Inorg. Chem.* **2003**, *42*, 7951-7958.
30. Wang, H.; Gordon, R. G.; Alvis, R.; Ulfig, R. M., Atomic layer deposition of ruthenium thin films from an amidinate precursor. *Chem. Vap. Deposition*, **2009**, *15*, 312-319.
31. Dreisch, K.; Andersson, C.; Stålhandske, C., Synthesis of MO_2Cl_2(N,N,N',N'-tetramethylethylenediamine) (M = Mo and W) and crystal structure of WO_2Cl_2(N,N,N',N'-tetramethylethylenediamine) —an unprecedented coordination geometry in the WO_2Cl_2 core. *Polyhedron* **1992**, *11*, 2143-2150.
32. Kamenar B.; Penavić, M.; Korpar-Čolig, B.; Marković, B., Preparation and crystal structure of two oxo-molybdenum complexes with dimethoxyethane. *Inorg. Chim. Acta,* **1982**, *65*, L245-L247.
33. Barry, S. T., Amidinates, guanidinates and iminopyrrolidinates: Understanding precursor thermolysis to design a better ligand. *Coord. Chem. Rev.*, **2013**, *257*, 3192-3201.
34. Ji, W.; Shen, R.; Yang, R.; Yu, G.; Guo, X.; Peng, L.; Ding, W., Partially nitrided molybdenum trioxide with promoted performance as an anode material for lithium-ion batteries. *J. Mater. Chem. A* **2014**, *2*, 699-704.
35. Pozo-Gonzalo, C.; Kartachova, O.; Torriero, A. A. J.; Howlett, P. C.; Glushenkov, A. M.; Fabijanic, D. M.; Chen, Y.; Poissonnet, S.; Forsyth, M., Nanoporous transition metal oxynitrides as catalysts for the oxygen reduction reaction. *Electrochim. Acta* **2013**, *103*, 151-160.
36. Kartachova, O.; Chen, Y.; Jones, R.; Chen, Y.; Zhang, H.; Glushenkov, A. M., Evolution of the electrochemical capacitance of transition metal oxynitrides with time: the effect of ageing and passivation. *J. Mater. Chem. A* **2014**, *2*, 12940-12951.

Table of Contents Graphic:



MOLYBDENUM OXO-AMIDINATE PRECURSOR
O1
O2
Mo1
N2
N2
N1
N1
^{1}H nmr
3.5
3.3
3.1
2.9
2.7
2.5
2.3
2.1
1.9
1.7
1.5
1.3
1.1
δ ppm
ALD GROWTH OF MoO_3 FILMS
MASS GAIN (ng/cm^2)
350
300
250
200
150
100
50
0
0
1000
2000
3000
4000
Time (s)